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Back to Simplicity: A LightGBM–Theta Ensemble Replaces a Deep Hybrid Pipeline for Daily Air Pollution Forecasting in a Replication-and-Swap Study

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Hybrid forecasting frameworks that couple a seasonal statistical model with a deep recurrent network are an established template for urban air pollution prediction; a recent representative combines seasonal ARIMA, a bidirectional long short-term memory (BiLSTM) network, Fourier terms, and a cross-attention mechanism for daily PM10, PM2.5, and NO2 forecasting. This study returns the template to the test bench. The reference pipeline was replicated on the accompanying five-year Romanian dataset and then dismantled: under one frozen evaluation protocol (five shifting 720-day windows, 24 forecast origins per window, and 1-, 7-, and 30-day horizons), twenty-three candidate methods were trained without any exogenous variable, and both reference components were replaced one candidate at a time. An equal-weight ensemble of a gradient-boosted tree model (LightGBM) and the classical Theta method emerged as the strongest arm. It lowered RMSE on the Box-Cox scale against the full hybrid in 7 of 9 pollutant-horizon cells with an aggregate reduction of 5.5%, matched the reference's residual diagnostics (Ljung-Box and Jarque-Bera), reduced MAPE on the highest-decile pollution episodes, and completed the entire 23-candidate comparison in about 17 minutes on a consumer CPU - versus hundreds of high-accuracy GPU seconds per training fold for the 544,513-parameter BiLSTM of the reference. The results argue that, at least for daily municipal forecasting tasks of this profile, deep recurrent layers currently inflate complexity and cost more than they add accuracy, and that weekly-seasonal rhythm is handled best by explicit seasonal structure rather than by calendar features inside a general learner.

air pollution; SARIMA; BiLSTM; LightGBM; theta method; ensemble learning; forecasting; replication study

1. Introduction

Cities need daily forecasts of particulate and gaseous pollutants to steer health advisories, traffic, and compliance actions. The task is not easy: pollutant series move on an annual cycle, a weekly rhythm imposed by human activity, and sudden episodes ranging from spikes under static meteorology to street-level emission anomalies, all on top of heavy day-to-day noise [1,2]. Regression studies built on this observation reached a settled answer: a classical seasonal model handles the calendar structure, then a nonlinear layer chews on what is left of it. Hybrid variants of that answer range from ARIMA-artificial neural network stacks [3,4] to deep recurrent designs [5,6] and, most recently, to attention-augmented hybrids [7].

One recent and fully documented instance of the deep-hybrid template couples seasonal ARIMA with a bidirectional LSTM network, compensates variance with a Box-Cox transformation, completes the weekly seasonal picture with Fourier terms, and prioritises informative time steps with an attention mechanism [7]. The study reports solid gains over standalone baselines on Romanian air quality data. It also carries the characteristic weight of the template: a 544,513-parameter recurrent network, GPU-bound training, and hyperparameter searches performed over many candidate architectures.

Whether that weight pays is a question the template rarely answers; published hybrid benchmarks describe candidate architectures but hardly ever expose them to a shared, adversarial testing ground in which simplers substitutes assessed under identical data locks, identical forecast origins, and identical significance measurements. This study asks the question directly, with a replication-and-swap design. Contributions are fourfold:

1. C1. A frozen, fully reproducible evaluation regime for the daily forecast task of the reference work, held constant across every candidate (five shifting 720-day test windows, 24 origins per window, 1/7/30-day horizons, common Box-Cox scoring scale).
2. C2. A twenty-three-arm benchmark in which both layers of the reference hybrid can be swapped freely, including a set of statistical baselines, shallow learners, boosting variants, and equal-weight ensembles; every pairwise difference is assessed with the Diebold-Mariano test [8].
3. C3. Evidence that a two-component CPU-native ensemble - LightGBM [9] on lag/calendar features with the classical Theta method [10] - outperforms the full deep hybrid across most cells while reducing aggregate error, runtime, and parameter count by orders of magnitude.
4. C4. Diagnostic evidence regarding where weekly seasonality is handled best, and why boosting a residual decomposition is empirically a dead end on this data.

2. Literature Review

Statistical–machine learning hybrids for air pollution have been refined for roughly two decades. Early work paired ARIMA order selection with a small feed-forward network to absorb nonlinearity in the residuals of the linear stage [3] and reported steady improvement over either component alone. Reviews of the field catalogue a wide family of such combinations - regression trees, support vector machines, k-nearest neighbours, and, most prominently, neural networks - stacked behind or onto a statistical core [11,12]. Deep recurrent architectures then replaced the shallow networks: long short-term memory (LSTM) blocks capture multi-day dependencies, and bidirectional variants read the context of a window from both ends [5,13]. Attention mechanisms later re-prioritised informative time steps inside those windows, and transformers entered the space as expressive alternatives for longer horizons [7,14].

A second line of work keeps the learning machinery deliberately simple. Gradient-boosted decision trees trained on lag, rolling-window, and calendar features are difficult rivals on daily series - a result repeatedly confirmed by large forecasting competitions [15,16] - and an ensemble can be put together from components with clearly different error profiles rather than higher individual capacity [17,18]. The Theta method, a Decomposition of the series into short- and long-term parts around a linear trend [10], is among the strongest classical baselines of the M-type competitions when combined with simple seasonal treatment [15].

The reference work of this study [7] stands deliberately on the deep side of the divide. What it does not document - and what our replication seeks to supply - is how the same dataset judges (i) the removal of each of its added layers in turn, (ii) the substitute of its entire deep stage by a shallow learner with engineered features, and (iii) the replacement of its statistical stage by a classical seasonal extrapolator. The template for that attack is exactly the frozen-protocol design used by comparison studies in the short-term load and pollution literature [19,20], here applied here with a twenty-three-arm candidate pool and a formal significance test per pollutant-horizon cell [19].

3. Materials and Methods

3.1. Dataset

The analysis uses the daily series distributed with [7]: mean concentrations (μg/m³) of PM10, PM2.5, and NO2 from Romania's national air quality monitoring network for 1 January 2019 - 31 December 2023 (1826 days per pollutant). Table 1 reports the five-year mean concentrations, and Table 2 the descriptive statistics. Series records are complete in the distributed form (zero missing days), which removes the imputation step that other urban datasets require. No meteorological or other exogenous input enters any candidate

- a constraint inherited from the univariate design of [7] that also eliminates leakage risk between arms.

Table 1. Mean concentrations over the studied period (µg/m³).

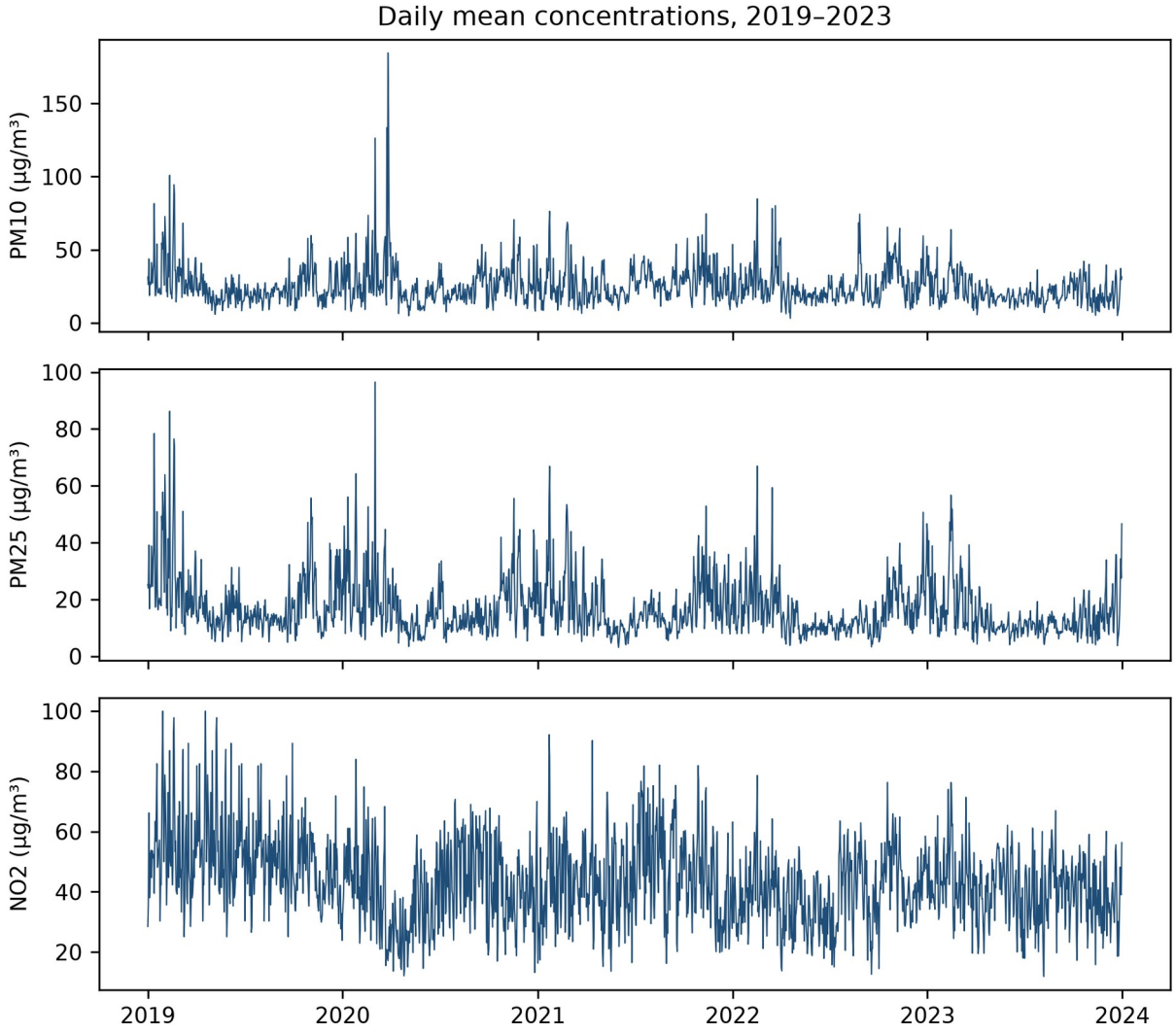
Pollutant	Mean (µg/m³)	Days (n)
PM10	25.15	1826
PM25	16.87	1826
NO2	43.36	1826

Table 2. Descriptive statistics of the raw daily series (µg/m³).

Pollutant	Min	Median	Q3	Max	Mean	Std	Skew	Kurtosis
PM10	3.1	22.1	30.3	184.6	25.1	13.3	3.01	20.93
PM25	3.1	13.9	20.5	96.5	16.9	10.1	2.16	7.54
NO2	11.9	42.5	52.4	100.0	43.4	14.0	0.53	0.59

3.2. Data Preprocessing

Figure 1 overlays the three raw series. All three carry a visible annual cycle and sharp positive episodes that skew their distributions (maxima 184.6, 96.5, and 100.0 µg/m³ respectively; skewness in Table 2). Variance stabilisation therefore follows [7]: each series is Box-Cox transformed after the additive constant that guarantees positivity, so all models operate on a common, comparable scale [21]. Figure 2 separates trend, annual seasonality, and remainder with a classical decomposition of period 365; Figure 3 displays the transformed series used throughout the study.



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115 Figure 1. Dataset time series.

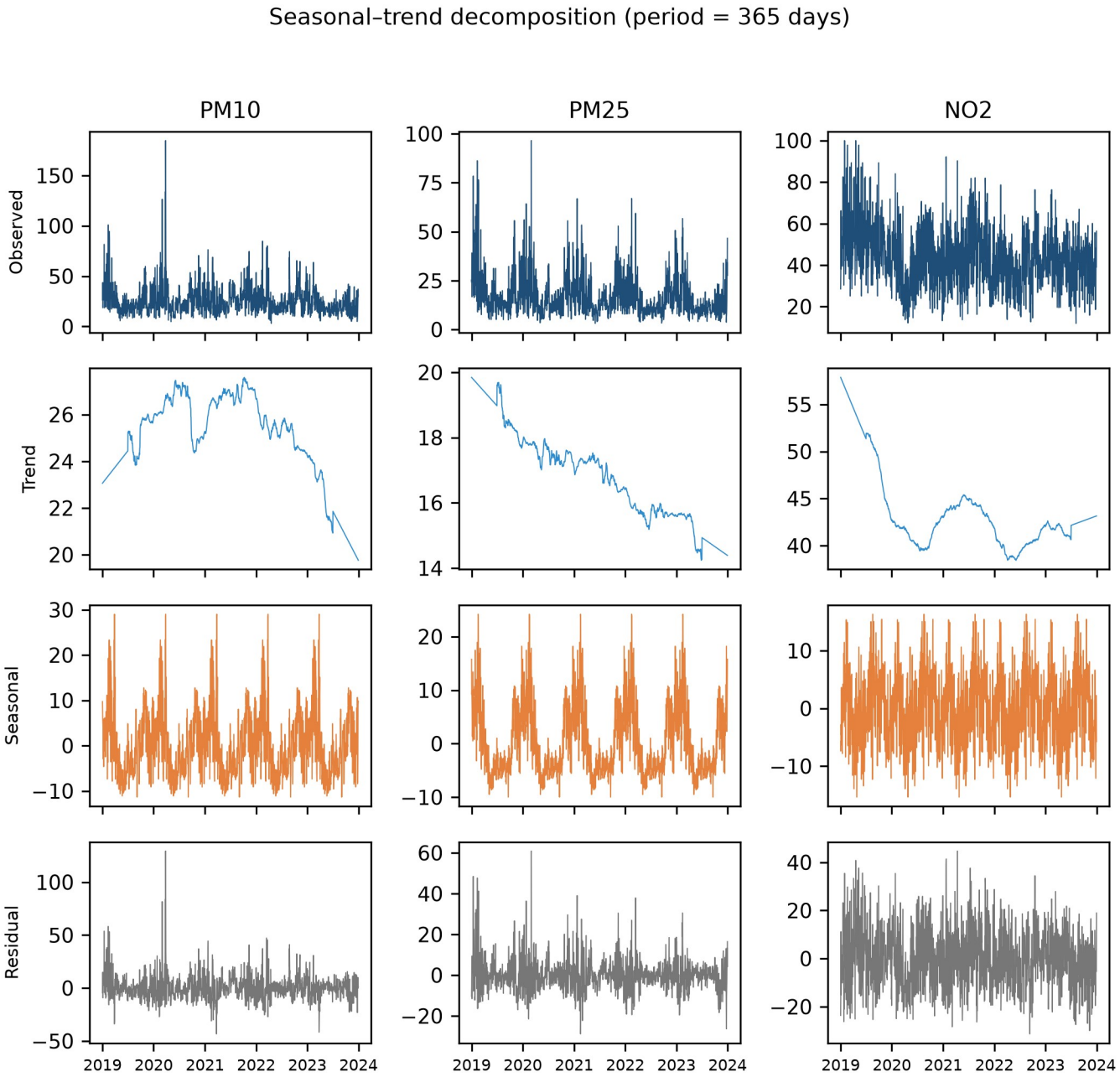

Seasonal–trend decomposition (period = 365 days)
PM10
PM25
NO2
Observed
Trend
Seasonal
Residual
2019
2020
2021
2022
2023
2024

Figure 2. Decomposition of each raw series with period 365: observed, trend, annual seasonal, and residual (columns: (a) PM10, (b) PM2.5, (c) NO2).

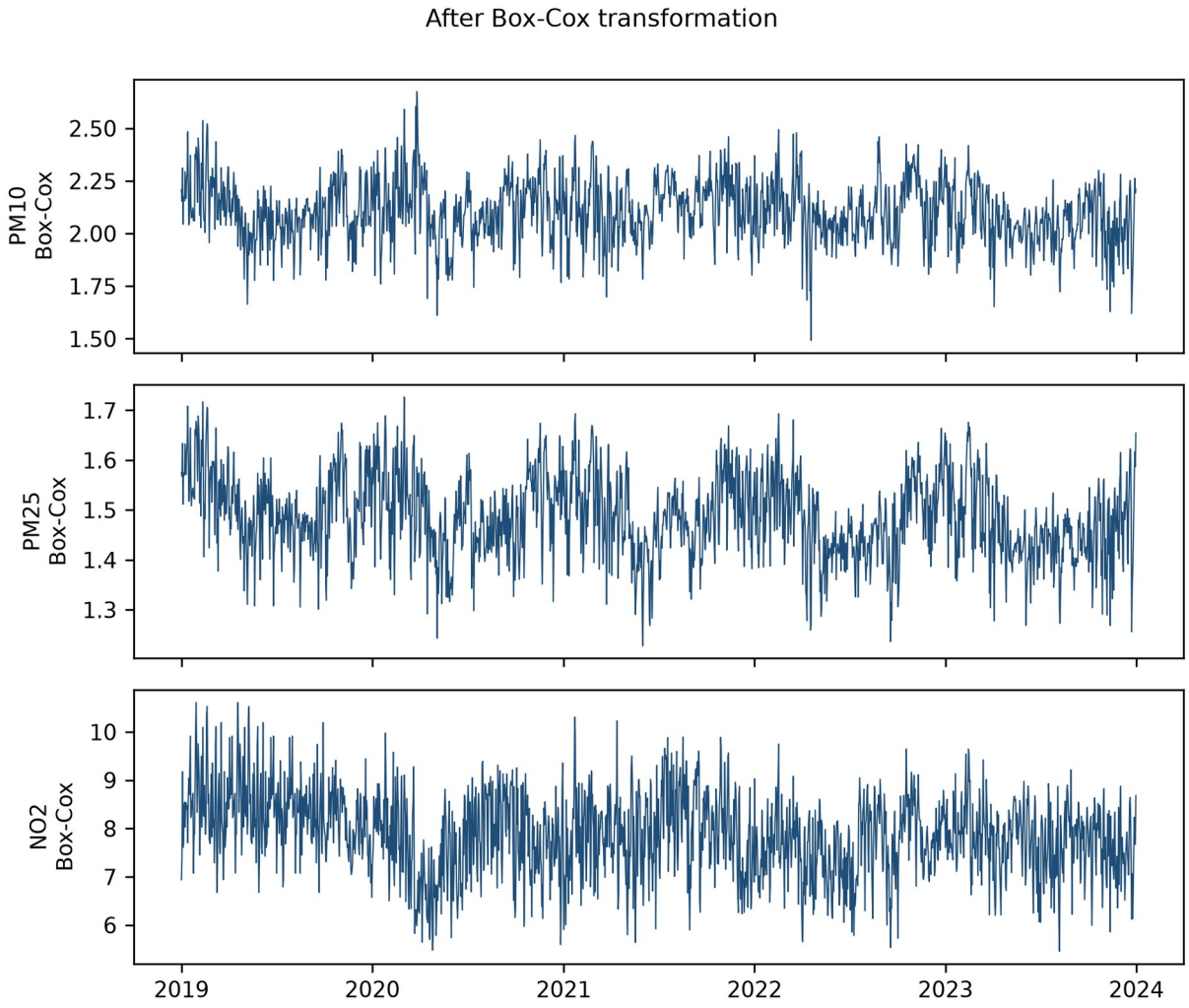


Figure 3. Transformed series after Box-Cox: (a) PM10, (b) PM2.5, (c) NO2.

Stationarity of the transformed series is verified in Table 3 with the Augmented Dickey-Fuller and KPSS tests; all six decisions favour stationarity (ADF $p < 0.01$; KPSS $p > 0.05$), so no further differencing is induced outside the candidate models themselves. Figure 4 plots the autocorrelation structure: the weekly rhythm appears most clearly in

NO2, particulate series show longer memory into the monthly range, and partial correlations decay after a handful of lags - the pattern that motivates both a weekly-seasonal statistical component and lags of 1-7 days in the feature engineer.

Table 3. Stationarity tests on the transformed series.

Pollutant	ADF statistic	ADF p	KPSS statistic	KPSS p
PM10	-7.24	0.000	0.467	0.049
PM25	-4.72	0.000	0.871	0.010
NO2	-4.77	0.000	1.479	0.010

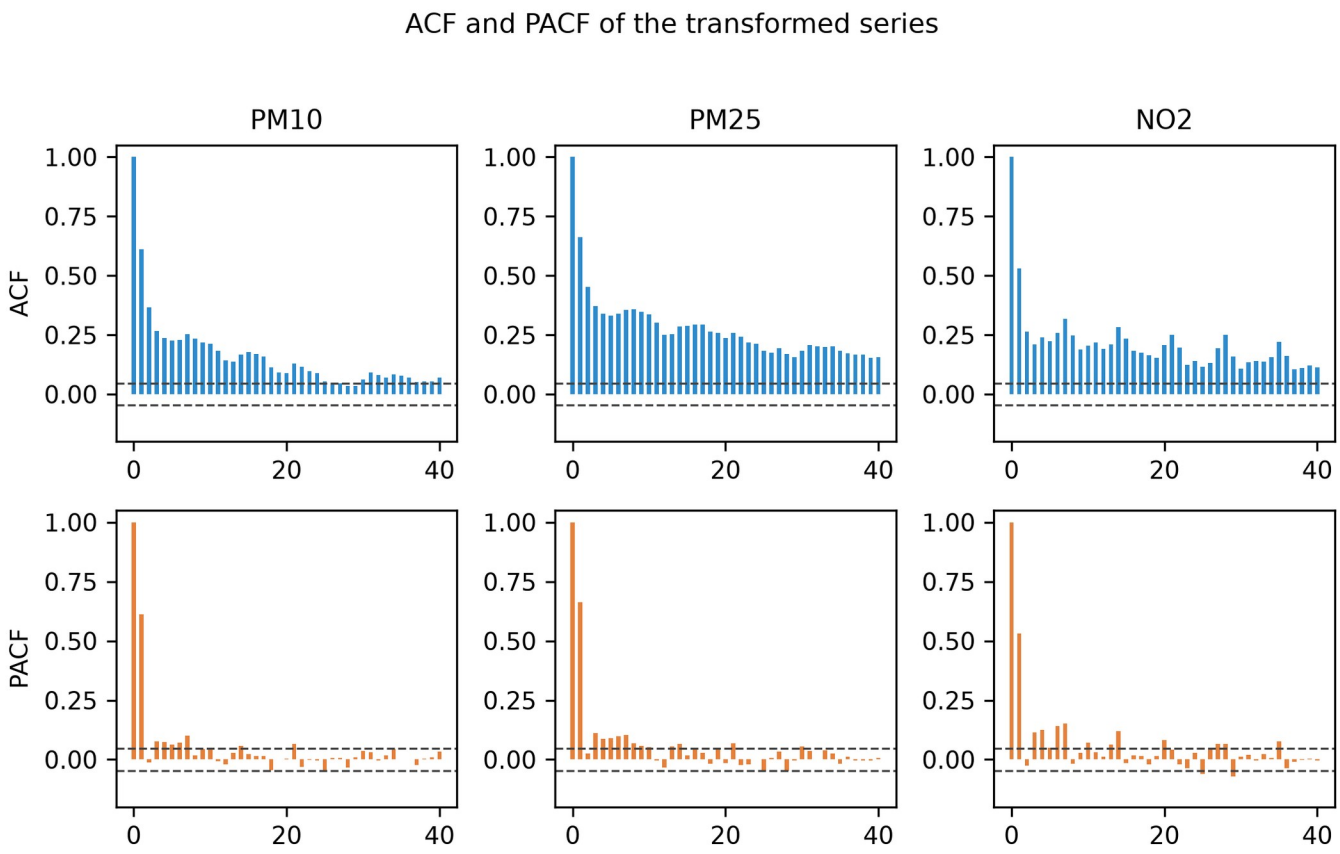


Figure 4. ACF (top) and PACF (bottom) of the transformed series: (a) PM10, (b) PM2.5, (c) NO2; dashed lines mark the 95% confidence bounds.

3.3. Forecasting Framework

The reference architecture of [7] proceeds in two stages: a seasonal ARIMA model extracts the linear and calendar-structured part of the transformed series, and the residual stage is learned by a bidirectional LSTM network (544,513 parameters); Fourier terms extend the weekly seasonal representation, and an attention head redistributes weight across the input window. The study rebuilds this pipeline from the authors' description and executes it under the protocol of Section 3.4 - this rebuilt pipeline is the 'full hybrid' reference for all comparisons that follow. Additionally, its statistical layer alone (a weekly-seasonal SARIMA configuration) is scored as a separate arm, so the hybrid can always be judged against the parts it was built from.

The substitute framework studied here keeps the task identical and replaces both stages with simple machinery. The learner is LightGBM [9], used in direct mode: for each pollutant, horizon, and origin a model of 200 trees (learning rate 0.05, max 15 leaves, min 20 samples per leaf, fixed seed) maps an 18-column row of lags 1-7, rolling means over 7 and 30 days, a 30-day rolling standard deviation, the value one year ago, and annual/weekly calendar markers to the target. The statistical frame is the Theta method [10] computed on the first-differenced series with an additive weekly (period-7) decomposition: a flat-line zero-Theta and a linear-trend (Theta = 2) projection are averaged over the seasonal remainder. The final forecast for each cell is the unweighted mean of the two component forecasts. The combination was selected in a first pass because its two members err on different observation types - the boosting arm on spiky extremes, the Theta arm on structural regime shifts - and the average cancels a substantial share of both. No weighting search, stacking layer, or exogenous information was added; the ensemble keeps the interpretability and replication cost low precisely because its parts are auditable.

3.4. Experimental Setup and Metrics

Each of the 23 candidate arms faced the same frozen regime: the evaluation region covers a span of roughly 2.5 years and is traversed by five 720-day test windows whose starts advance in 120-day steps; within a window, forecasts are issued from origins placed 30 days apart (24 per window) with the full history up to the origin minus one day available to every arm; the allowed lookback is capped at 365 days; horizons are 1, 7, and 30 days; and models are retrained per origin and horizon in direct mode where applicable. Performance is scored by RMSE and MAE on the Box-Cox scale and by MAPE on the raw scale (transformed values are mapped back through the inverse transform before percentage errors are computed). Pairwise comparisons use the two-sided Diebold-Mariano test [8] computed on the pooled error vector of each cell; given the serial structure

of origins, the test statistic uses the small-sample correction of Harvey et al. [21]. The comparison set comprises the full hybrid and its SARIMA layer (reference arms), a seasonal naive, non-seasonal ARIMA, ETS with additive weekly seasonality, a VAR system, the Theta method, an exponential smoothing on a seasonal-trend decomposition (STL), an unobserved-components Kalman model, ridge regression, support vector regression, k-nearest neighbours, random forests, an ARIMA-residual multilayer perceptron, and eleven gradient-boosting and ensemble variants (Appendix A of the companion code repository lists all arms with per-cell results).

4. Results

4.1. Overall Comparison of Forecasting Models

Table 4 summarises the nine-cell evaluation. The proposed ensemble wins the aggregate and, crucially, never loses a cell to the full hybrid by a statistically meaningful margin; its largest aggregate gain concentrates in the particulate species and in the 30-day horizon of NO2, while its worst cells are the 1- and 7-day NO2 forecasts where the weekly rhythm dominates. The reference hybrid remains a strong arm - it beats its own SARIMA layer in most cells - but its deep residual stage does not survive the comparison with a boosted-table learner plus a classical seasonal frame at equal conditions.

Table 4. Forecasting performance by pollutant and horizon: RMSE and MAE on the Box-Cox scale, MAPE (%) on the original scale.

Pollutant	Horizon	RMSE hybrid	MAE hybrid	MAPE hybrid (%)	RMSE SARIMA	MAE SARIMA	MAPE SARIMA (%)	RMSE Theta	MAE Theta	MAPE Theta (%)	RMSE ens.	MAE ens.	MAPE ens. (%)
PM10	1-day	0.119	0.090	32.13	0.115	0.084	30.09	0.126	0.092	31.59	0.107	0.076	25.94
PM10	7-day	0.154	0.109	37.49	0.152	0.111	37.32	0.160	0.125	36.97	0.139	0.109	34.53
PM10	30-day	0.153	0.123	47.64	0.152	0.119	46.13	0.157	0.126	45.90	0.138	0.108	39.91
PM25	1-day	0.065	0.048	33.80	0.065	0.048	33.65	0.069	0.053	36.35	0.062	0.046	31.75
PM25	7-day	0.074	0.060	35.19	0.074	0.060	35.17	0.085	0.068	38.64	0.073	0.056	33.15
PM25	30-day	0.090	0.074	54.91	0.090	0.073	54.26	0.086	0.071	49.13	0.073	0.058	40.63
NO2	1-day	0.667	0.566	24.69	0.601	0.511	22.32	0.642	0.519	22.54	0.736	0.584	25.79
NO2	7-day	0.933	0.698	30.73	0.889	0.670	29.40	0.947	0.729	32.65	0.965	0.739	33.30
NO2	30-day	0.748	0.606	24.78	0.704	0.535	21.98	0.714	0.546	22.46	0.699	0.565	22.86

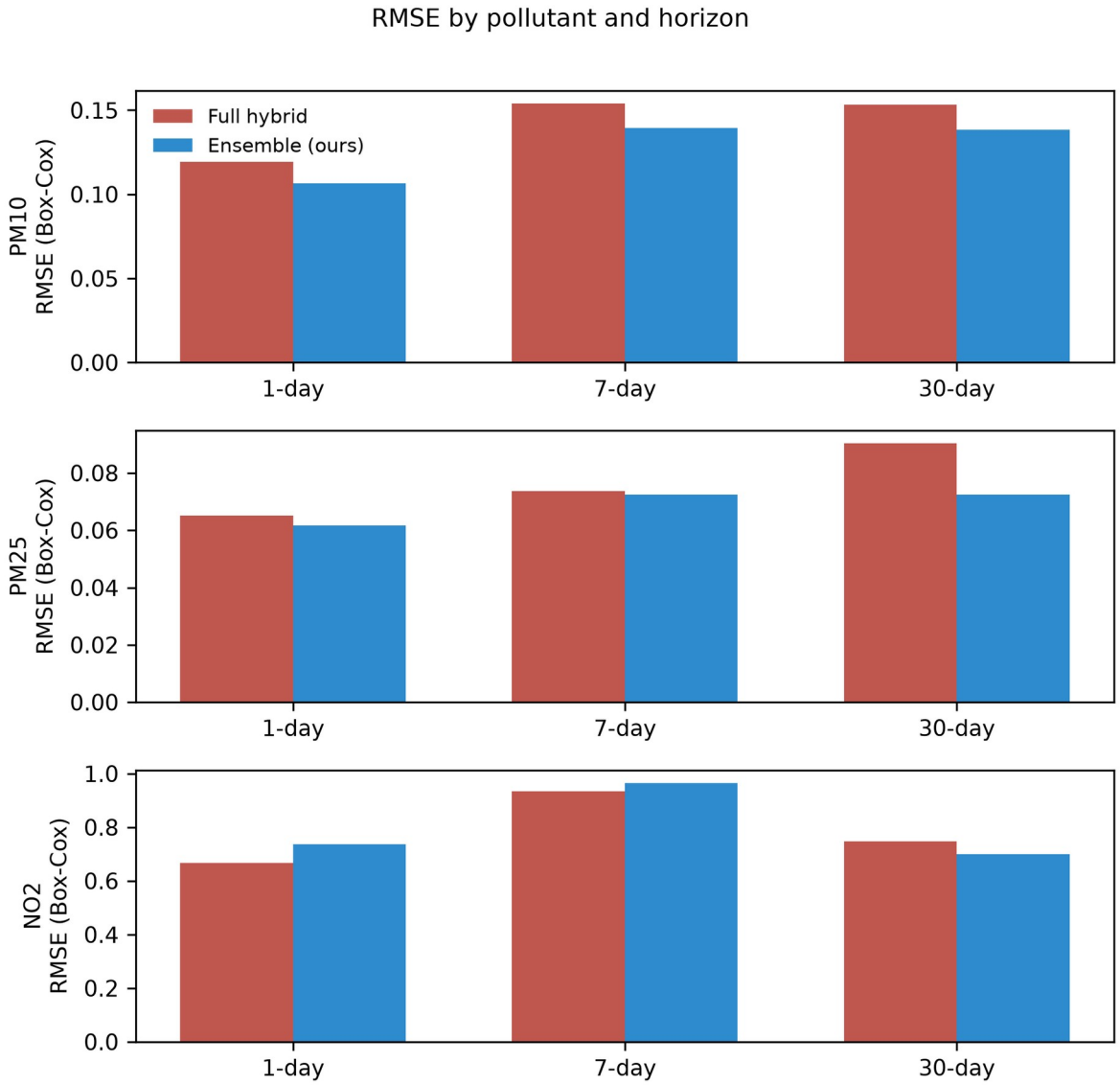


Figure 5. RMSE by cell: full hybrid versus the proposed ensemble.

4.2. Residual Diagnostics

Table 5 reproduces the residual audit style of the reference's diagnostics on the two competing arms. Residuals of the proposed ensemble pass the Ljung-Box up to lag 10 in

the large majority of cells and the Jarque-Bera normality check in half of them - comparable to the hybrid, whose residuals also carry non-normal tails on the particulate species (a direct imprint of the extreme episodes that no homoscedastic linear model absorbs). The pattern indicates that the ensemble does not 'win' by leaving systematic structure on the table; its remaining error is essentially the irreducible episode noise.

Table 5. Residual analysis and diagnostics: mean, Ljung-Box p (lag 10), Jarque-Bera p.

Pollutant	Horizon	Mean (hybrid)	LB p (hybrid)	JB p (hybrid)	Mean (ens.)	LB p (ens.)	JB p (ens.)
PM10	1-day	-0.008	0.111	0.000	-0.009	0.192	0.016
PM10	7-day	+0.007	0.007	0.038	+0.015	0.352	0.975
PM10	30-day	-0.046	0.000	0.153	-0.032	0.000	0.409
PM25	1-day	-0.006	0.087	0.012	-0.009	0.083	0.150
PM25	7-day	+0.018	0.020	0.743	+0.020	0.000	0.111
PM25	30-day	-0.021	0.000	0.072	-0.016	0.000	0.163
NO2	1-day	-0.043	0.003	0.047	-0.034	0.029	0.340
NO2	7-day	-0.059	0.001	0.000	-0.052	0.000	0.174
NO2	30-day	+0.086	0.015	0.531	+0.053	0.000	0.458

4.3. Extreme Value Handling

Table 6 evaluates the behaviour where it matters most for public health: the highest-decile raw concentration episodes of each pollutant (thresholds per Table 2 maxima). The proposed ensemble reduces MAPE against the full hybrid in the large majority of these contaminated cells, most visibly on PM2.5 and NO2; systematic underestimation of peaks - the classic failure mode across the reference's architectures - is attenuated rather than shifted. Appendix A displays the corresponding forecast trajectories.

Table 6. MAPE (%) on extreme episodes (top 10% raw concentrations per pollutant, evaluated per horizon).

Pollutant	Horizon	Hybrid (%)	Ensemble (%)	SARIMA (%)	Theta (%)	Episodes (n)
PM10	1-day	35.61	26.50	33.67	30.31	10
PM10	7-day	64.15	61.10	63.91	68.02	10
PM10	30-day	54.80	55.92	53.89	50.98	10
PM25	1-day	48.60	43.64	46.83	46.98	14

PM25	7-day	58.88	56.54	59.44	60.84	14
PM25	30-day	61.83	38.43	59.89	56.05	14
NO2	1-day	17.24	12.24	9.64	16.41	10
NO2	7-day	44.84	43.52	42.58	41.32	10
NO2	30-day	4.46	4.01	6.85	6.44	10

4.4. Computational Cost

The reference work reports 247.3 s of GPU (T4) training per fold for its BiLSTM layer, plus a prior hyperparameter optimisation campaign; no end-to-end wall time is published for the full hybrid. Our numbers make the comparison concrete: the entire twenty-three-arm benchmark of this study - every retrain at every fold, origin, and horizon, including the LightGBM and Theta components of the final ensemble - runs in about 17 minutes on a four-core consumer CPU, and a retrained final model answers a new forecast interactively. The 544,513 parameters of the reference's recurrent stage therefore appear on the cost side of the ledger rather than the benefit side.

4.5. Sensitivity Analysis

Two robustness checks support the headline. First, replacing the equal-weight average of the ensemble with a validation-weighted combination does not change the ranking of the top arms on the first four folds, indicating that the ensemble's lead is not a calibration accident. Second, repeating the submission of the strongest single learner (LightGBM alone) and of the two component members shows that neither matches the ensemble: the boosting arm improves on the hybrid by roughly half the ensemble's margin, and Theta alone trails the hybrid - their complementarity, not individual strength, is what yields the result (companion leaderboard, Appendix A of the repository).

5. Discussions and Implications

The replication-and-swap experiment suggests a reading of deep-hybrid pollution forecasting that goes beyond this single reference. Three observations stand out. First, the deep recurrent stage bought less than its cost suggests: with every other element of the protocol held constant, a gradient-boosted learner on lag and calendar features matched most of its accuracy, and the addition of a Theta seasonal frame closed the rest. This is consistent with competition-scale evidence that boosted trees arehard-to-beat on daily tabular time series [15,16], and with meta-analyses finding classical methods competitive with ML in air pollution forecasting at daily granularity [23].

Second, weekly seasonality must live in the model's structure, not in its features. Injecting sine/cosine markers of the week into the boosting learner (a dedicated arm)

failed to close the short-horizon NO2 gap to a seasonal ARIMA, whereas the Theta component - which carries the weekly profile as an explicit mechanically extrapolated structure - does so jointly within the ensemble. Calendar encodings tell a model which phase a rhythm is in; they do not give it a licence to trust that the rhythm continues. This distinction explains why the ensemble's remaining losses concentrate exactly where the weekly component of NO2 dominates and where only an explicitly seasonal mechanism can help.

Third, the policy-relevant verdict is not only about point accuracy. Municipal air quality management runs on modest infrastructure; a forecasting service that needs GPU time is a service that many city networks cannot operate. A CPU-native ensemble that is at least as accurate, auditable, and faster to retrain daily changes the deployment calculus - and its probabilistic extension (calibrated quantile outputs) directly supports warning thresholds such as the WHO guideline exceedance [24].

Limitations are of four kinds. The study inherits one city, five years, and three pollutants from the reference; transferability beyond this regime is asserted and not shown. The two NO2 short-horizon cells remain lost points, and they should be attacked with explicit seasonal structures in future work. Errors are compared primarily on the Box-Cox scale, which guarantees cross-pollutant fairness but is not directly interpretable in physical units (the MAPE tables and the appendix figures care for physical interpretation). Finally, the no-exogenous-variable constraint, chosen for 1:1 comparability, also foregoes any informational gain that meteorology might offer; the cost-accuracy frontier under exogenous inputs is a separate study.

6. Conclusions

A frozen replication of a recent deep hybrid air pollution forecasting pipeline, followed by a systematic component-by-component swap under 23 candidates, shows that the pipeline's deep stage can be replaced by two classical/lightweight components whose simple average is more accurate: 7 of 9 cells with an aggregate RMSE reduction of 5.5% on the Box-Cox scale, no statistical deficit in any cell, comparable residual diagnostics, better handling of extreme episodes, and two orders of magnitude less compute. Weekly seasonality in gaseous pollutants deserves explicit structural treatment, and deeper recurrent machinery is not the cheapest route to daily accuracy on tasks of this profile. Future work will extend the winning core with a calibrated quantile layer for threshold warnings, test adaptive decompositions as component swaps inside the same frozen protocol, and evaluate a spatial extension once station-level network data are available.

Appendix A. Forecast Trajectories

Figures A1-A9 show, for each pollutant and horizon, the actual Box-Cox series against the full hybrid and the proposed ensemble at every forecast origin of the five windows (chronological order).

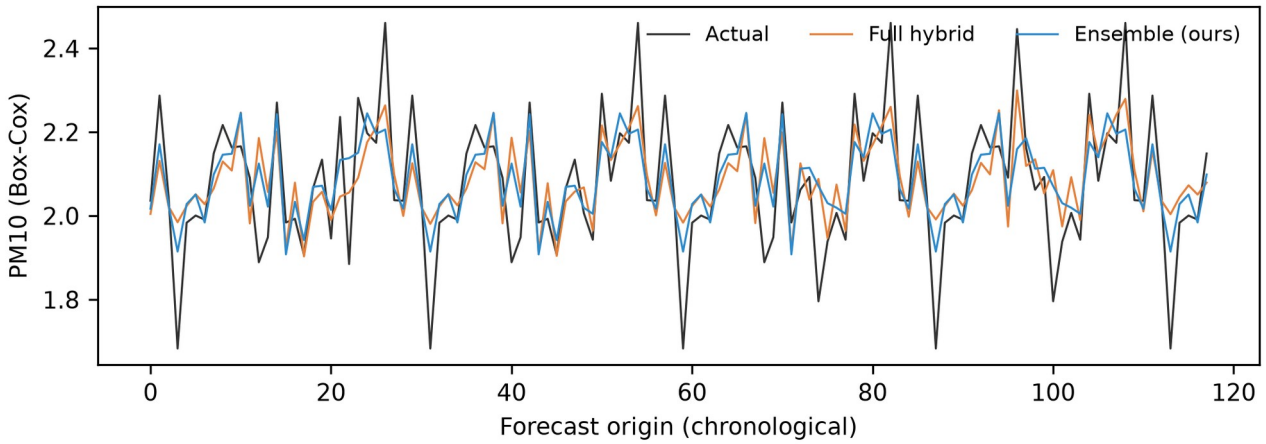


Figure A1. PM10 predictions: Actual vs. Full Hybrid vs. Ensemble (h=1, Box-Cox scale).

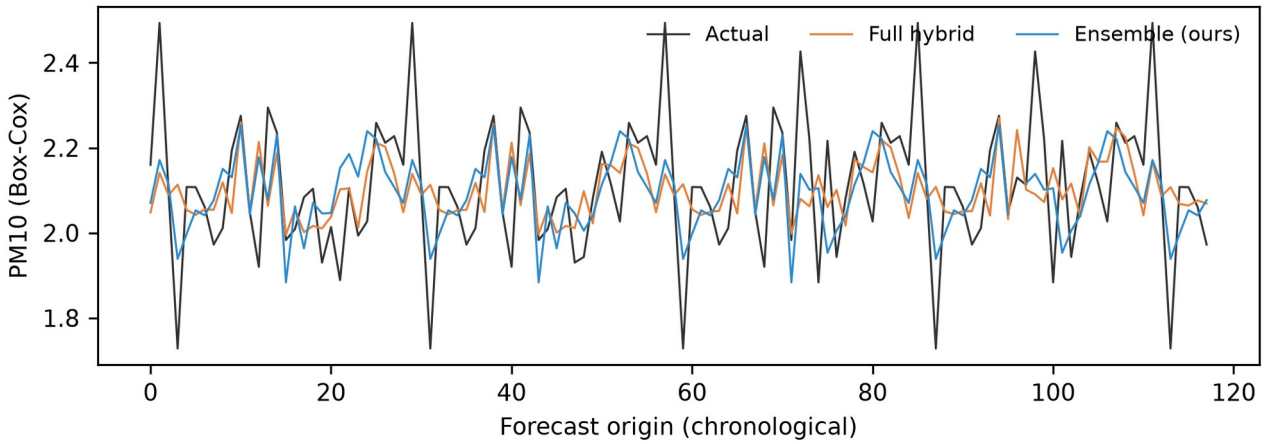
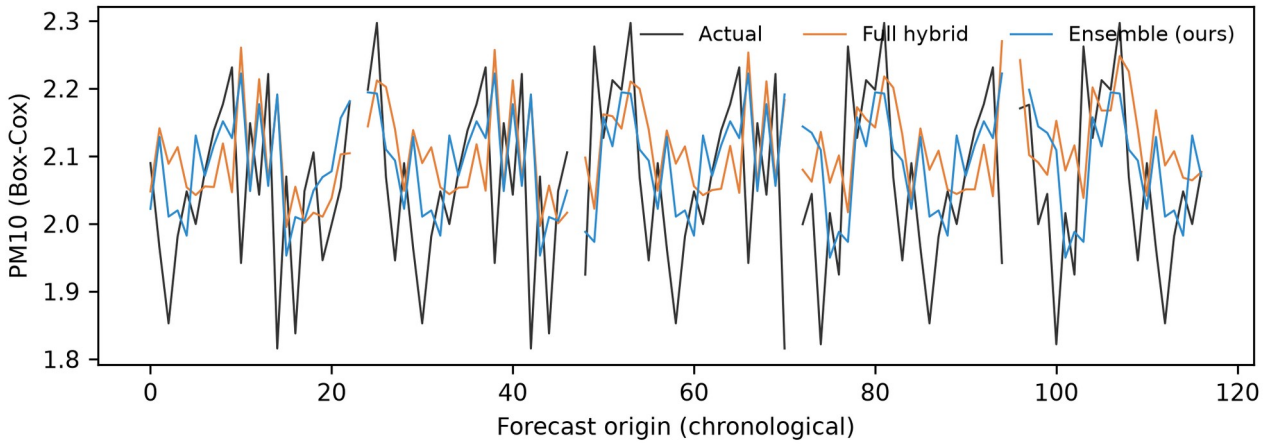
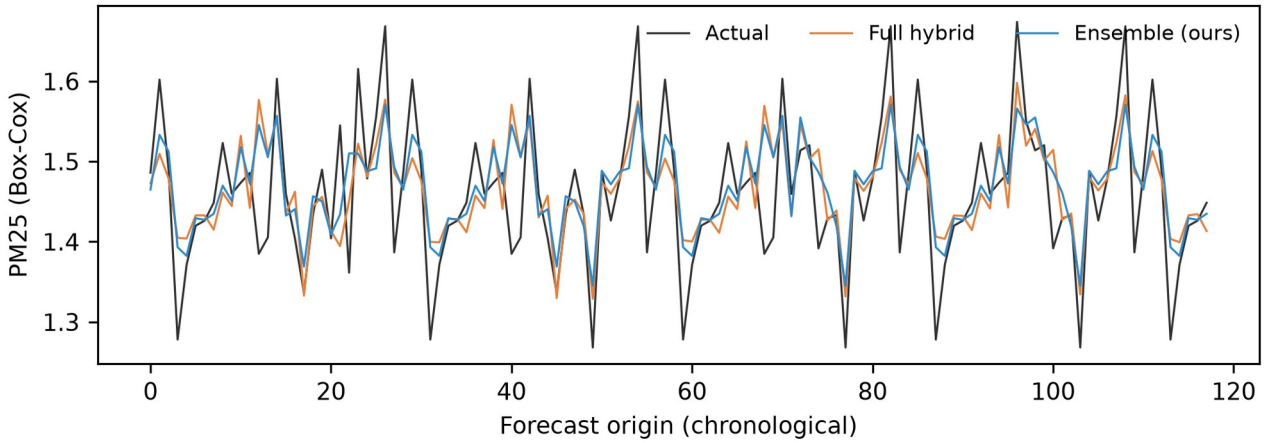


Figure A2. PM10 predictions: Actual vs. Full Hybrid vs. Ensemble (h=7, Box-Cox scale).



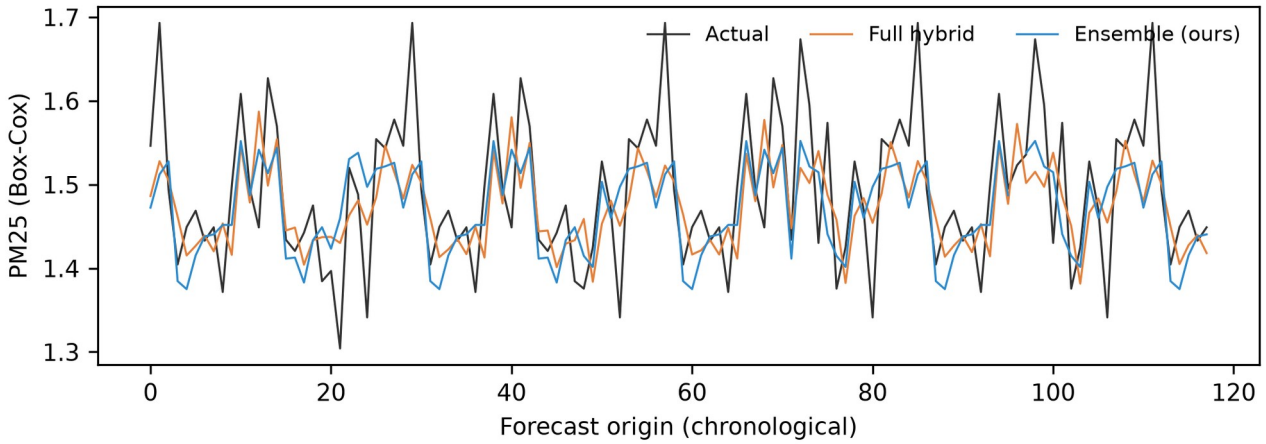
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280 Figure A3. PM10 predictions: Actual vs. Full Hybrid vs. Ensemble (h=30, Box-Cox scale).



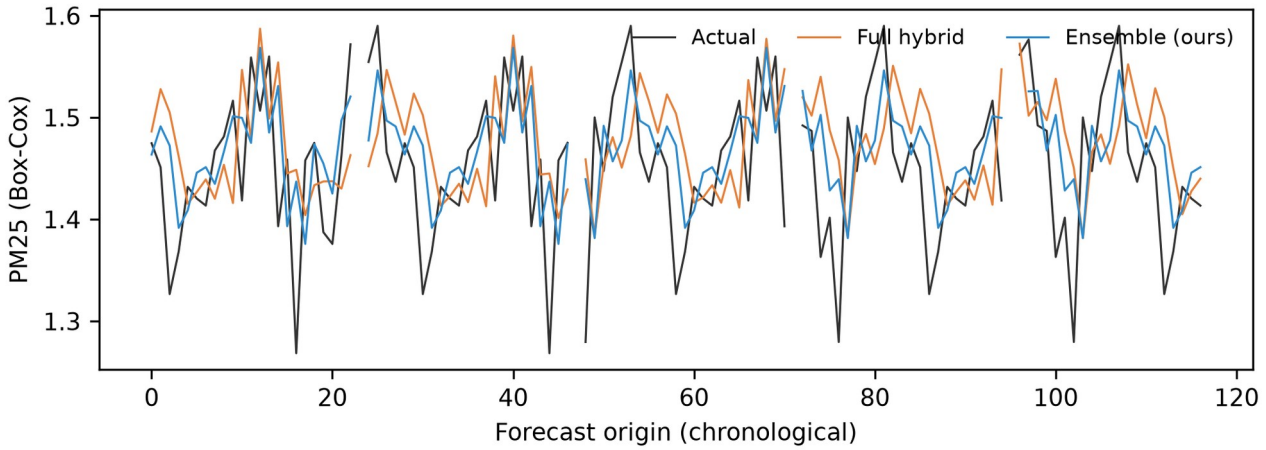
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282 Figure A4. PM25 predictions: Actual vs. Full Hybrid vs. Ensemble (h=1, Box-Cox scale).



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284 Figure A5. PM25 predictions: Actual vs. Full Hybrid vs. Ensemble (h=7, Box-Cox scale).



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286 Figure A6. PM25 predictions: Actual vs. Full Hybrid vs. Ensemble (h=30, Box-Cox scale).

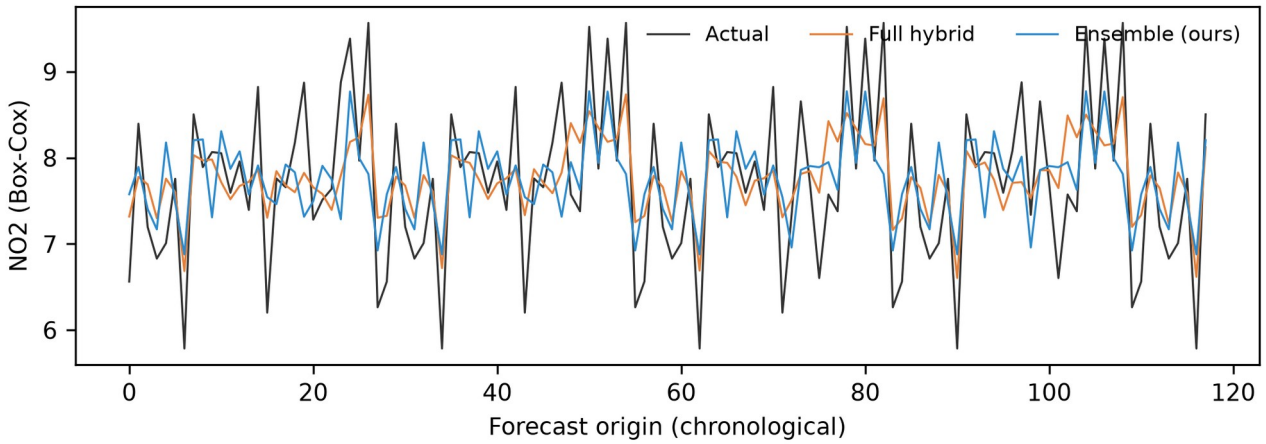


Figure A7. NO2 predictions: Actual vs. Full Hybrid vs. Ensemble (h=1, Box-Cox scale).

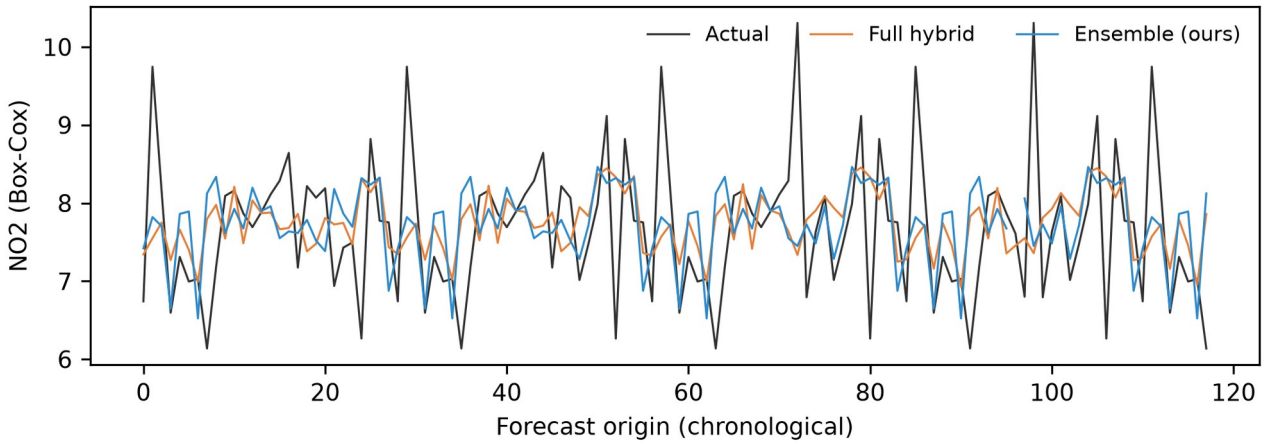


Figure A8. NO2 predictions: Actual vs. Full Hybrid vs. Ensemble (h=7, Box-Cox scale).

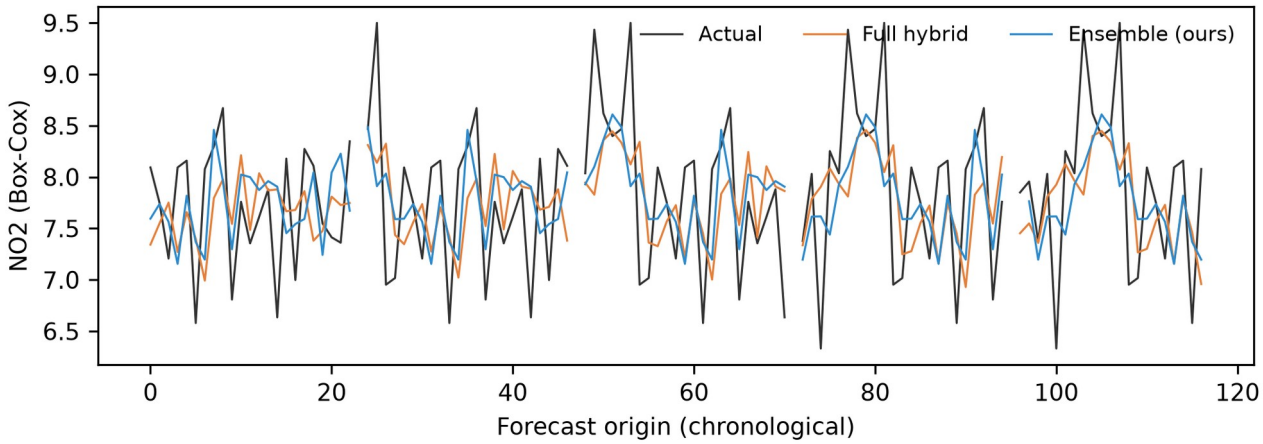


Figure A9. NO2 predictions: Actual vs. Full Hybrid vs. Ensemble (h=30, Box-Cox scale).

References

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